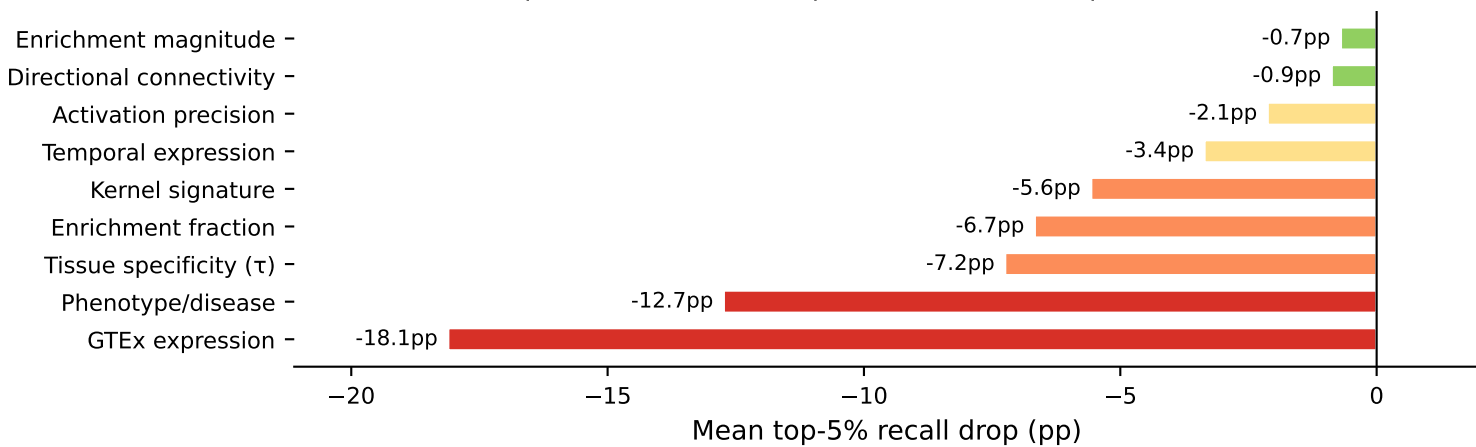
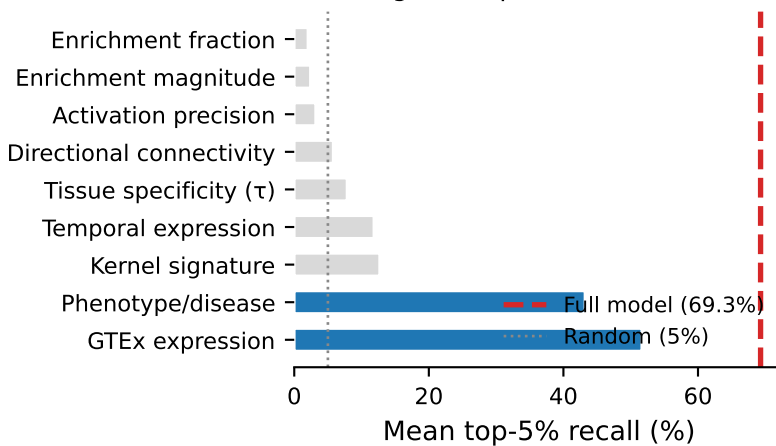


a

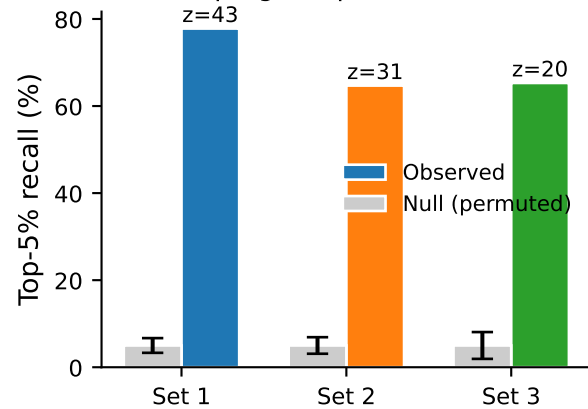
Component ablation — impact when each component removed

**b**

Single-component baselines

**c**

Gene-program permutation null

**d**

Propensity-matched null: each real factor paired with a TF matched on GTEx expression level and phenotype annotation density

